

Density-Assisted Evolutionary Dynamic Multimodal Optimization

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Dynamic Multimodal Optimization Problems (DMMOPs) demand algorithms capable of swiftly locating and tracking multiple optimal solutions over time. The primary challenge lies in controlling the population diversity to facilitate effective exploration, all within the limitation of computational resources between consecutive environmental changes. In this article, we study the utilization of density information derived from both current and historical populations to enhance exploration. First, for each active sub-population, we construct a density landscape based on the distribution of concurrently active sub-populations and establish dominance relationships between candidate solutions in the sub-population based on density and fitness values, directing this sub-population toward exploring low-density promising areas. Then, for each converged sub-population, we construct a density landscape based on the distribution of sub-populations that have historically become extinct, guiding the restart of this sub-population in low-density unexploited areas. Finally, we develop a comprehensive framework of Density-Assisted Evolutionary Algorithm (DAEA), which encompasses density-assisted search and restart, also combined with initialization. Moreover, we employ prediction and memory strategies to enhance the performance of DAEA in dynamic environments. Notably, the algorithm relies on an external monitor to detect environmental changes and trigger the dynamic response strategy. DAEA is tested on the CEC'2022 dynamic multimodal optimization benchmark suite and is compared against several state-of-the-art dynamic multimodal optimization algorithms. The experimental results demonstrate the competitiveness of DAEA in handling DMMOPs. Additionally, experimental results from the berth allocation problem further confirm the applicability of DAEA to real-world dynamic multimodal optimization tasks.

CCS Concepts: • **Theory of computation** → **Evolutionary algorithms**;

Additional Key Words and Phrases: Dynamic multimodal optimization, Evolution strategy, Kernel density estimation, Prediction

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1 Introduction

Many real-world optimization problems have multiple global optima and/or acceptable local optima, such as protein structure prediction [Wong et al., 2010], multi-project scheduling [Pérez et al., 2016], and automatic itinerary planning [Huang et al., 2020]. These problems are recognized as **Multimodal Optimization Problems (MMOPs)**, and identifying all global optima and acceptable local optima can provide decision-makers with valuable insights into problem understanding and candidate solution selection [Li et al., 2016]. However, the unavoidable dynamic environment in many optimization problems increases the difficulty of problem-solving. Take city navigation as an example, it could propose several viable parking options for drivers, but urban traffic networks which undergo constant dynamic changes necessitate the optimizer to consistently update multiple acceptable optimal paths in changing environments [Lin et al., 2022a; Zheng and Geroliminis, 2016]. Therefore, **Dynamic Multimodal Optimization Problems (DMMOPs)** require optimization algorithms to rapidly explore all optimal solutions and track them in every environmental context. More specifically, the key to addressing a DMMOP is to tackle its multimodality and dynamics [Lin et al., 2022b].

Evolutionary Algorithms (EAs), such as genetic algorithm [Cheng et al., 2018a; Wang et al., 2015], differential evolution [Jiang et al., 2023; Lin et al., 2021], and **Evolution Strategy (ES)** [Luo et al., 2022b; Xu et al., 2021c], have been widely employed to solve MMOPs and have achieved favorable results. Typically, conventional EAs tend to converge to a single area, requiring the incorporation of supplementary strategies to assist their exploration of multiple promising areas. Niching methods are a representative and popular class of multimodal optimization strategy, guiding the population to explore various promising areas explicitly or implicitly [Lin et al., 2022b]. Among these methods, such as crowding [Mahfoud, 1992], speciation [Pérowski, 1996], and **Nearest-Better Clustering (NBC)** [Preuss, 2010], sub-populations are frequently formed based on the relative spatial position among individuals. Notably, Xu et al. [2021b,c] investigated how dividing the population based on a well-distributed initial population can lead to improved and more stable results. Moreover, some approaches leverage information from temporal gaps in sub-population evolution to motivate sub-populations to search uncharted areas. For example, Wei et al. [2022] penalize the sub-populations that fall into previously explored areas, and Ahrari et al. [2022a, b] prevent sub-populations from conducting repetitive searches within the same area by defining a repelling radius for each sub-population.

Once the environment changes, the fitness values acquired from prior environments become outdated. Although it is possible to sustain the evolutionary process by reevaluating the population, the loss of population diversity reduces its exploration efficiency after an environmental change [Nguyen et al., 2012]. Moreover, treating the problem following an environmental change as a completely new one is limited by computational resources available between successive environmental changes [Yazdani et al., 2021]. Therefore, dynamic response strategies constitute integral components of dynamic optimization algorithms [Lin et al., 2022b]. For example, strategies include restarting entire or partial population to control diversity [Kazemi Kordestani et al., 2020; Yazdani et al., 2020], creating the explicit archives for storing historical solutions and retrieving them after environmental changes [Cao et al., 2018; Vafashoar and Meybodi, 2020], adapting the

computational resource allocation to different components [Yazdani et al., 2023b], and acquiring knowledge of patterns in successive environments to predict the optimal solution and expedite the search following environmental changes [Liu et al., 2018; Liu et al., 2020; Meier and Kramer, 2020]. Additionally, numerous experiments have demonstrated that the niching method is also positive strategy in the context of dynamic optimization [Bu et al., 2017; Li et al., 2023; Lung and Dumitrescu, 2007], and its potential has been extensively explored.

Because of the remarkable performance of niching methods in both multimodal and dynamic optimization, dynamic multimodal optimization algorithms, from the inception of traceable DMM-CSA [Luo et al., 2019], have consistently integrated niching methods as essential components to maintain the population diversity [Ahrari et al., 2022a, b; Cuevas et al., 2021] and then complemented by dynamic response strategies, such as prediction, which serve to address the issues arising from the dynamic environment [Ahrari et al., 2021; Wu et al., 2023]. However, to efficiently search for and track multiple moving optimal peaks, it is necessary to control the global diversity of the population throughout the evolutionary process, specifically the coverage of promising areas by multiple sub-populations and the computational resource consumption of converged sub-populations [Yazdani et al., 2021].

To tackle the challenge of global diversity loss on DMMOPs, this study proposes a density model to assist EAs, called **Density-Assisted Evolutionary Algorithm (DAEA)**. DAEA employs **Kernel Density Estimation (KDE)** [Wand and Jones, 1994] to create a density landscape that corresponds to the current and/or historical population. It drives sub-populations to cover as much of the low-density areas as possible and restarts the converged sub-populations in low-density areas. Moreover, the prediction and memory strategies are incorporated to enhance the performance of DAEA in dynamic environments. Notably, DAEA does not include an active environmental change detector. Instead, it operates under the assumption that environmental changes occur when the allocated fitness evaluations are exhausted for the current system state [Luo et al., 2022a]. In summary, our contributions are as follows:

- (1) We provide a complete framework of DAEA that encompasses population initialization, search, and restart, which are assisted by a density model in the different evolutionary stages to maintain the global diversity of the population. First, density-assisted initialization is employed to obtain an initial population that balances uniformity and randomness [Xu et al., 2021c]. Second, as a sub-population evolves, the density landscape constructed by other concurrently evolving sub-populations is adopted, and the dominance relationships between candidate solutions are defined based on the density and fitness value to drive this sub-population to explore low-density and better fitness value areas. Third, as sub-populations converge, the density landscape constructed by historically extinct sub-populations is adopted to assist in restarting sub-populations in low-density unexploited areas.
- (2) We employ the prediction method and memory strategy to assist in generating the initial population and learning the knowledge of the new environmental fitness landscape. First, the **Long Short-Term Memory (LSTM)** [Hochreiter and Schmidhuber, 1997] is used to estimate fitness landscapes after environmental changes, drawing on the information of the historical environmental initial population, which could reduce the substantial fitness evaluation cost required for the density-assisted initialization. Then, the memory strategy is used to accelerate the process of the initial population in locating high-quality solutions within changing environments.

DAEA is tested on the CEC'2022 dynamic multimodal optimization benchmark suite [Luo et al., 2022a]. For comparative analysis, top-ranked algorithms participating in the CEC'2022 and CEC'2023 competition on seeking multiple optima in dynamic environments are employed.

The experimental results suggest that DAEA exhibits a high level of competitiveness. Additionally, DAEA is tested on a real-world application, the **Berth Allocation Problem (BAP)**, where the results show its superiority in both the quality and quantity of the solutions found.

The rest of this article is organized as follows. Section 2 introduces the related work. Section 3 describes the details of the proposed algorithm. Section 4 shows the experimental design and results. Section 5 provides the ablation study and analyzes dynamic response strategy and parameters of DAEA. Section 6 demonstrates the performance of DAEA in real application problems, i.e., BAPs. Finally, Section 7 concludes this article and introduces future work.

2 Related Work

In this section, we first define and summarize the progress of research on DMMOPs and then introduce KDE and long- and short-term memory.

2.1 Dynamic Multimodal Optimization

Recently, significant attention has been directed toward investigating dynamic optimization problems and MMOPs [Xiong et al., 2023; Zhen et al., 2023]. For example, Gao et al. [2023] utilized distributed knowledge transfer by employing sub-group cross-matching and self-adjusting knowledge transfer probability strategies to effectively transfer local knowledge between different tasks, enhancing overall performance across all tasks in solving MMOPs. Wang et al. [2023] proposed a condition knowledge representation and feedback learning framework based on contrastive reinforcement learning to address the uncertainty in the state space of dynamic optimization problems. Employing dynamic optimization models and Monte Carlo policy gradient learning architectures effectively enhances the performance of the solutions. However, limited attention has been devoted to simultaneously addressing dynamic and multimodal challenges, namely DMMOPs, which encompass problems with multiple global and/or local optimal solutions, and where the problem environment dynamically changes. This concept was first thoroughly discussed by Nanas and De Roeck [2007]. Cheng et al. [2018b] first provided the basic design of DMMOPs by the combination of dynamic optimization problems and MMOPs. Without loss of generality, a maximization DMMOP that considers only global optima takes the following form:

$$\Phi_t = \arg \max_{\mathbf{x} \in \Omega} f(\mathbf{x}, t), \quad (1)$$

where $\Phi_t = \{\mathbf{x}_1^*, \dots, \mathbf{x}_{n_t}^*\}$ represents the set of n_t global optima of objective function $f(\mathbf{x}, t)$ in t th environment, and Ω is the decision space. In this article, we consider DMMOPs only with boundary constraints and assume that the decision space Ω does not change over time.

Cheng et al. [2018b] tested a variant of brain storm optimization, called BSO-OS, on a DMMOP. Then, Luo et al. [2019] developed the first dynamic multimodal optimization algorithm, known as DMMCSA. It elucidated the fundamental components of evolutionary multimodal optimization algorithms, including the optimizer, niching method, and dynamic response strategy. Endeavors of subsequent studies have predominantly designed dynamic multimodal optimization algorithms based on those components. Regarding niching methods, Cuevas et al. [2021] proposed a modified mean shift algorithm, which combines density estimation and gradient ascent to locate optimal solutions. Ahrari et al. [2022a, b] presented an improved variant of RS-CMSA-ES [Ahrari et al., 2017], i.e., RS-CMSA-ESII. This variant controls the taboo radius and searches for each sub-population in a more sophisticated way. Regarding dynamic response strategies, Ahrari et al. [2021] proposed an adaptive multilevel prediction method, designed with distinct prediction levels. In this study, lower-level predictions could assimilate knowledge from the higher-level predictions to enhance accuracy. Wu et al. [2023] proposed a **Kriging Model-Based Evolutionary Algorithm with a**

Support Vector Machine (KMEA-SVM), which identifies promising solutions using the Kriging model and predicts solution quality for new environments adopting support vector machines.

2.2 KDE

KDE [Wand and Jones, 1994] is a non-parametric method to estimate the probability density of an unknown sample point based on a given sample set. Assuming a set of n independent and identically distributed sample points $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$, the density function $\rho(\mathbf{x})$ is as follows:

$$\rho(\mathbf{x}) = \frac{1}{N} \sum_{i=1}^N K(\mathbf{x} - \mathbf{x}_i), \quad (2)$$

where $K(\cdot)$ is the non-negative kernel function, and the Gaussian kernel function stands out as one of the most extensively employed options. Equation (3) shows a variant of Gaussian kernel function [Xu et al., 2021c].

$$K(\mathbf{x} - \mathbf{x}_i) = \begin{cases} \exp(-\|\mathbf{x} - \mathbf{x}_i\|^2/2h^2), & \text{if } \|\mathbf{x} - \mathbf{x}_i\| \leq h, \\ 0, & \text{else.} \end{cases} \quad (3)$$

In the domain of machine learning, KDE is often used as an important tool in clustering methods. Recent research has employed KDE to enhance the performance of EAs. For example, Li et al. [2023] adopted a fast density peak clustering as a niching method to create multiple sub-populations. Cuevas et al. [2021] proposed a modified mean shift technique, utilizing a density attractor to identify local maximum density weighted based on their fitness value. Moreover, Xu et al. [2021b] proposed a **Density-Based Population Initialization Strategy (DPIS)** for generating a population that balances uniformity and randomness. Subsequent enhancements, namely **Improved DPIS (IDPIS)** and **Fitness-Weighted DPIS (FDPIS)**, were proposed [Xu et al., 2021c]. IDPIS confines the density influence range of individual sample points, as shown in Equation (3), whereas FDPIS employs fitness values to weigh the density space, called FKDE, aiming to uncover potential global peaks. In Xu et al. [2021c], IDPIS and FDPIS are called successively with the pseudo-code shown in Algorithm 1.

2.3 LSTM

The LSTM [Hochreiter and Schmidhuber, 1997] is a well-known variant of **Recurrent Neural Network (RNN)** specifically designed for processing and learning of time series data. It aims to address the vanishing and exploding gradient problems encountered in RNN when learning long-term dependencies. In comparison to traditional RNNs, LSTM demonstrates superior capability in retaining and conveying information when handling extended sequences.

The LSTM unit incorporates a cell c and three gates, namely forget gate f , input gate i , and output gate o . Each gate contains a sigmoid activation function to control the flow of information. Moreover, LSTM has a cell state, and the cellular state allows information to flow continuously through the sequence, independent of the gating unit. The structure of LSTM is shown in Figure 1, and the details are as follows, where W_f, W_i, W_c, W_o represent the weights, $\mathbf{b}_f, \mathbf{b}_i, \mathbf{b}_c, \mathbf{b}_o$ represent the biases, σ and $\tanh$ denote the sigmoid and hyperbolic tangent functions, respectively, and $\odot$ represents the Hadamard product [Hochreiter and Schmidhuber, 1997].

Initially, at the time t , LSTM incorporates the prior timestep output h_{t-1} and cell state c_{t-1} , both from time $t - 1$, and integrates them with input vector $\mathbf{x}_t$ at the current time in the network. Then, the forget gate f_t and input gate i_t within LSTM collaboratively dictate the cell state at time t . The forget gate decides which information from the preceding state c_{t-1} is to be discarded, while the input gate determines which information from the current input $\mathbf{x}_t$ is preserved and stored

Algorithm 1: Density-Assisted Initialization [Xu et al., 2021c]

Input: The population size generated by IDPIS N_1 , the population size generated by FDPIS N_2 ,
Output: The initial population $\mathcal{P}$

- 1 $\mathcal{P}_1 \leftarrow \emptyset$;
- 2 Randomly generate the first individual $\mathbf{x}_1$;
- 3 $\mathcal{P}_1 \leftarrow \mathcal{P}_1 \cup \{\mathbf{x}_1\}$;
- 4 **for** $i \leftarrow 2, \dots, N_1$ **do**
- 5 Randomly sample k positions $\{\mathbf{x}'_1, \dots, \mathbf{x}'_k\}$;
- 6 $\{\rho_1, \dots, \rho_k\} \leftarrow \text{KDE}\{\mathbf{x}'_1, \dots, \mathbf{x}'_k\}$;
- 7 Select a position with the lowest ρ as $\mathbf{x}_i$;
- 8 $\mathcal{P}_1 \leftarrow \mathcal{P}_1 \cup \{\mathbf{x}_i\}$;
- 9 **end**
- 10 Evaluate the population $\mathcal{P}_1$;
- 11 Randomly sample m position;
- 12 $\{\rho_{f_1}, \dots, \rho_{f_m}\} \leftarrow \text{FKDE}\{\mathbf{x}'_1, \dots, \mathbf{x}'_m\}$;
- 13 Select N_2 positions with the highest ρ_f as $\mathcal{P}_2$;
- 14 $\mathcal{P} \leftarrow \mathcal{P}_1 \cup \mathcal{P}_2$;
- 15 **Return** $\mathcal{P}$

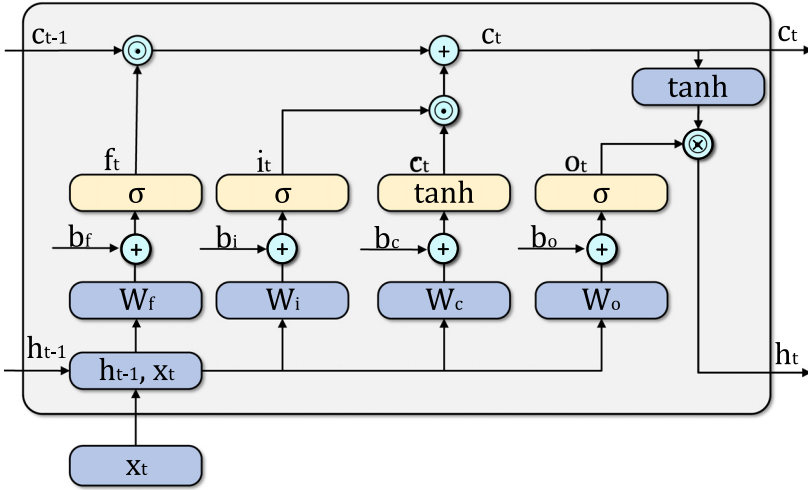


Fig. 1. The schematic of LSTM.

[Hochreiter and Schmidhuber, 1997].

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f), \quad (4)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i). \quad (5)$$

LSTM proceeds by calculating the cell state $\tilde{c}_t$ in correspondence with the current input vector x_t , subsequently updating the cell state c_t at the current network.

$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c), \quad (6)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t. \quad (7)$$

Finally, the information flow reaches the output gate o_t , where it selectively outputs useful information from the cell state c_t . The output gate enables LSTM to preserve valuable long-term dependencies, essential for predictions at both the current and future timesteps [Hochreiter and Schmidhuber, 1997].

$$o_t = \sigma(W_o \cdot [\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o), \quad (8)$$

$$h_t = o_t \odot \tanh(c_t). \quad (9)$$

3 The Proposed Algorithm

In this section, we first discuss and analyze the challenges faced by existing dynamic multimodal optimization algorithms, providing illustrative examples to highlight the motivation behind the development of our proposed algorithm. We then present the comprehensive framework of the DAEA, detailing its key components: density-assisted population initialization, search, and restart mechanisms, and dynamic response strategies, including prediction and memory techniques. Finally, we analyze the time complexity of DAEA and summarize its similarities and differences with other algorithms of the same type, underscoring the advantages of DAEA.

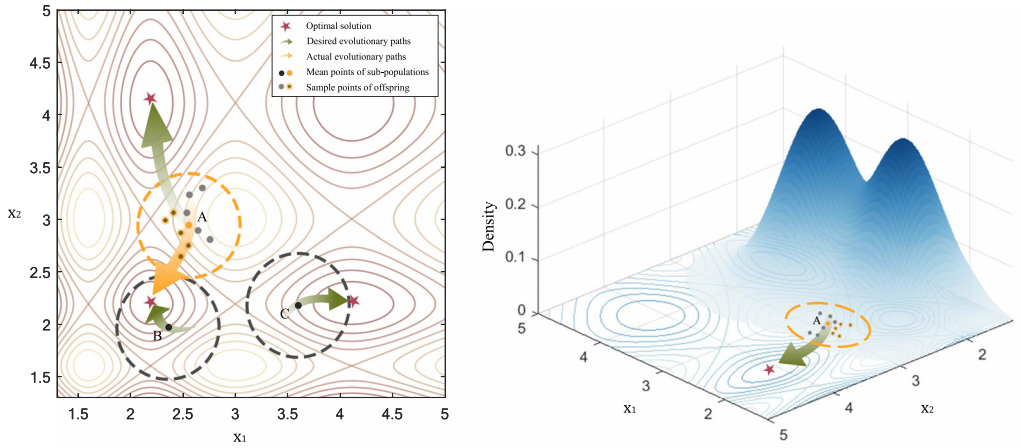
3.1 Motivation

In dynamic multimodal optimization, a key challenge is how to quickly search multiple global optima and/or acceptable local optima on request in each environment [Lin et al., 2022b], and niching methods have been widely acknowledged and validated through various studies [Li et al., 2016], which assist EAs in simultaneous exploration of diverse potential regions by dividing the population into multiple sub-populations. Nevertheless, the risk of the niching methods is that different sub-populations could redundantly search the same area without external intervention. This redundancy results in global diversity loss and inefficient utilization of computational resources.

Figure 2(a) takes the function F_7 from the CEC'2013 multimodal optimization benchmark and uses CMA-ES [Hansen and Ostermeier, 2001] as the optimizer to illustrate the possible global diversity loss in niching methods. In the initialization stage, the population is divided into three sub-populations A , B , and C , each intended to explore separate global optimal peaks, as indicated by the green arrows representing the desired search directions. However, after the first sampling of sub-population A , its evolutionary trajectory might deviate from the expected path during the evolutionary process, as illustrated by the orange arrow. Eventually, sub-populations A and B end up exploring the same area repeatedly without discovering the desired optimal solution, leading to a global diversity loss.

There have been many efforts to mitigate this risk to more efficiently utilize computational resources. For example, Ahrari et al. [2017, 2022a, b] proposed the implementation of a taboo area for each sub-population, which prevents other sub-populations from exploring the same area by maintaining a taboo distance between them. Wei et al. [2022] used a penalty strategy to penalize individuals in the neighborhood area of elite solutions and guide them toward unexplored areas. However, when multiple sub-populations search simultaneously, it is a challenge to use the distribution of concurrently active sub-populations to steer the exploration of the current sub-population in the direction of increasing global diversity.

To address this challenge, we revisit the evolution of sub-populations from the perspective of density space. As shown in Figure 2(b), all sub-populations except the evolved one unite to form a density space, where the explored or being explored areas are represented as the high-density areas, and unexplored areas are represented as the low-density areas. Therefore, besides individuals with



(a) Evolutionary path of three sub-populations. (b) Density peaks formed sub-populations B and C.

Fig. 2. (a) Evolutionary case of three sub-populations exploring globally optimal peaks in a local area, where the glowing orange points are selected to generate the next mean of the search distribution. (b) Density peaks are formed by concurrently active sub-populations when sub-population A is explored, where the desired evolutionary path of sub-population A is pointing to an area with lower density.

better fitness values, individuals in low-density areas are equally deserving of attention to increase global diversity and avoid the waste of computational resources caused by repeated sampling.

Following this raw idea, first, we describe the proposed DAEA for multimodal optimization in the rest of this section, and then, a prediction mechanism and memory strategy are introduced to enhance the performance of the DAEA in a dynamic environment. Finally, we provide a complete framework of DAEA.

3.2 DAEA

EAs usually start by randomly generating a set of candidate solutions and continuously iterating the population by evolutionary operators to find the high-quality solutions. Moreover, some additional strategies, such as restarting, are often employed to increase the global diversity of the population when it converges prematurely. In this section, we describe in detail how the density model assists those three main stages of EAs, i.e., initialization, search, and restart, to enhance the performance of EA.

Stage I—Initialization. The first stage of EAs is to randomly sample a set of points as initial candidate solutions. Xu et al. [2021b,c] proposed to employ a density model to assist in constructing an initial population, in which candidate solutions are generated in the low-density areas to take into account both uniformity and randomness of the initial population. In this article, this approach is adopted, specifically described in Section 2.2 and Algorithm 1.

Stage II—Search. The search stage of EAs improves the quality of the candidate solutions by evolving the population continuously to find the global high-quality solution. In this stage, the efficient use of computational resources to hold the exploratory efficiency of the population is crucial. For this reason, an intuitive idea is that the population should search unexplored areas as much as possible and avoid repetitive searches in explored areas. With this purpose in mind, a density model is employed to assist the search during evolution to enhance the exploration efficiency of the population.

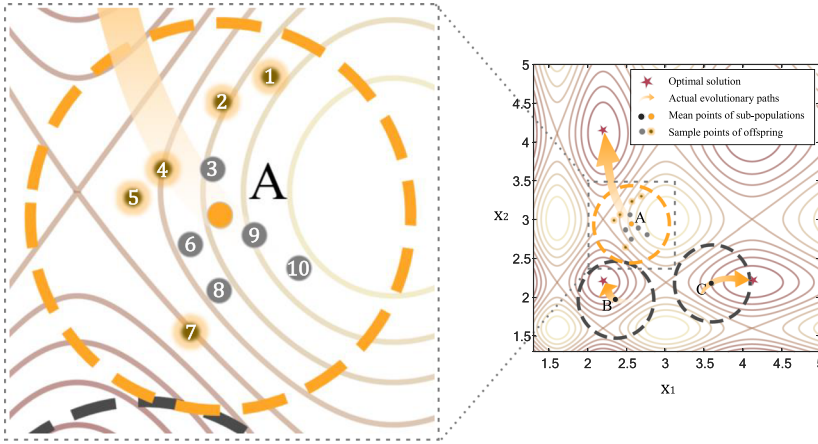


Fig. 3. Evolutionary case of three sub-populations exploring global optimal peaks in a local area with the assistance of the density model, where the glowing orange points are used to generate the next mean of the search distribution.

In the search stage, a niching-based multimodal EA expects each sub-population to explore a potential area separately to find as many global high-quality solutions as possible, and the coverage area of a sub-population could represent its exploration area. Assuming that there are several sub-populations sorted by best fitness value in descending order $\mathcal{P} = \{P_1, \dots, P_n\}$ in i th generation, then for sub-population P_j , the coverage area of sub-populations $\{P_1, \dots, P_{j-1}\}$ and sub-populations that have converged to global optimal solutions should reject the search of sub-population P_j . To characterize this area more precisely, we use the KDE to quantify the effect of this area in density space on the search, as shown in Equations (2) and (3). To find high-quality solutions while preserving global diversity, we redefine comparison for individuals in terms of dominance relations.

Definition 1. For two solutions $\mathbf{x}, \mathbf{x}' \in \Omega$ with the pair of fitness value and density $[f(\mathbf{x}), \rho(\mathbf{x})]$, $[f(\mathbf{x}'), \rho(\mathbf{x}')]$, $\mathbf{x}$ dominates $\mathbf{x}'$ (i.e., $\mathbf{x}$ is better than $\mathbf{x}'$, denoted by $\mathbf{x} \succ \mathbf{x}'$), if $f(\mathbf{x}) \geq f(\mathbf{x}')$ and $\rho(\mathbf{x}) \leq \rho(\mathbf{x}')$, and at least one of the inequalities is strict: $f(\mathbf{x}) > f(\mathbf{x}')$ or $\rho(\mathbf{x}) < \rho(\mathbf{x}')$. Additionally, if $\rho(\mathbf{x}) = 0$ and $\rho(\mathbf{x}') > 0$, $\mathbf{x} \succ \mathbf{x}'$.

For a set of candidate solutions, the nondominated sorting approach [Deb et al., 2002] is used to create different nondomination levels. There is a slight difference here that the solutions $\mathbf{x}$ with $\rho(\mathbf{x}) = 0$ are first placed in the first level because they could increase global diversity in the low-density area. After this, solutions at the same level are further sorted according to fitness value to obtain the final ranking.

Figure 3 displays a different situation resulting from the same example as Figure 2(a) but using the nondominated sorting, where solutions $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_4, \mathbf{x}_5, \mathbf{x}_7$ are nondominated solutions in the first level, solutions $\mathbf{x}_5 \succ \mathbf{x}_6, \mathbf{x}_8$, solutions $\mathbf{x}_2 \succ \mathbf{x}_3$, and solutions $\mathbf{x}_1 \succ \mathbf{x}_9, \mathbf{x}_{10}$. Compared with the case of Figure 2(a), solutions $\mathbf{x}_1, \mathbf{x}_2$ instead of solutions $\mathbf{x}_6, \mathbf{x}_8$ are selected and recombined together with $\mathbf{x}_4, \mathbf{x}_5, \mathbf{x}_7$ to generate a new mean. With the assistance of the density model, the sub-population A is driven to search toward the lower-density promising global area.

However, not all sub-populations are promising to find globally high-quality solutions, and ineffective exploration wastes a lot of computational resources. Hence, drawing from the termination criteria proposed by Maree et al. [2018], which include re-exploring niches and converging to a

local optimum, we tailor two judgment conditions for DAEA. These conditions aim to identify sub-populations with inefficient search and promptly eliminate them to conserve computational resources. The first condition is set to detect whether the sub-population repeats the search for the explored area. When the maximum cumulative number of high-density samples monitored for a sub-population reaches the threshold, it is inferred that the sub-population redundantly searches the high-density area, i.e., an explored area, which results in the global diversity loss and inefficient utilization of computational resources, leading to its elimination. The threshold $\varepsilon_{\text{hd},i}$ of each sub-population P_i is inversely proportional to the initial average distance between it and the other sub-populations, defined as follows:

$$\varepsilon_{\text{hd},i} = \alpha \cdot \left(\frac{1}{n} \sum_{j=1}^n \|\bar{\mathbf{x}}_i - \bar{\mathbf{x}}_j\|_2 \right)^{-1}, \quad (10)$$

where $\bar{\mathbf{x}}$ represents the mean point of the sub-population, and α represents the scale factor. This is because the larger the initial average distance between P_i and the other sub-populations, the lower the possibility that different initial populations will explore the same region, i.e., the threshold of the maximum cumulative number of high-density samples monitored for a sub-population threshold should be lower.

The second condition is set to detect whether the sub-population converges to local areas. First, we use the evolutionary step σ of the sub-population to determine the state of the sub-population, which is considered to be close to convergence if the evolutionary step σ is less than a threshold ε_σ . Then, we use the difference between the approximate global optimal solution and the optimal solution of the current sub-population to determine the area of the sub-population, which is considered to be located in a local optimal area when the sub-population optimal value does not reach the global optimal value. Here, we adopt a relaxed approach to identify local optima, acknowledging them only when the upper bounds on this sub-population optimal value and the global optimal value differ, i.e., $\lceil f(\mathbf{x}_i^*) \rceil \neq \lceil f(\mathbf{x}^*) \rceil$, where $\mathbf{x}_i^*$ is the optimal solution of the sub-population and $\mathbf{x}^*$ is the best solution found so far.

The pseudocode of the second stage of DAEA, i.e., density-assisted search, is displayed in Algorithm 2, which is also detailed as follows:

DAEA additionally calculates the densities of sample points during the offspring generation of sub-population P_i , and high-density sampling point counter c adds the number of densities greater than 0 in P_i . In this step, the density model is created by the concurrently better sub-populations and the sub-populations that have converged. Then, the rankings of sample points are obtained using nondominated sorting instead of fitness value-based sorting. Next, the sub-population is sequentially judged whether it explores a high-density area or converges a local area and returns the empty set to indicate that the sub-population is discarded. Note that the first condition is that the high-density sampling point counter c exceeds the $\varepsilon_{\text{hd},i}$. The second condition determines whether the sub-population has converged to a local area only after $\sigma < \varepsilon_\sigma$. Finally, the sub-population is updated according to the optimizer used, and this process is repeated for I_{max} iterations.

Stage III—Restart. The restart strategy is an intuitive way used in EAs to enhance global diversity by reinitializing the population. Similar to the search stage, the restarted population should be encouraged to explore new areas whenever possible. Therefore, we incorporate the restart mechanism into the stage of DAEA and use a density model to assist in restarting the population.

In the search stage, DAEA records the exploration trajectory of each sub-population as $\mathcal{T}_j$. For sub-population P_j , its exploration trajectory consists of continuously moving mean points $\mathcal{T}_j = \{\bar{\mathbf{x}}_1, \bar{\mathbf{x}}_2, \dots\}$. Then, the exploration trajectories of all historically extinct sub-populations

Algorithm 2: Density-Assisted Search

```

1 Input: Current sub-population  $P_i$ , current iterations  $\mathcal{I}_{current}$ , maximum iterations per cycle
    $\mathcal{I}_{max}$ , high-density sampling point counter  $c = 0$ , potential global optimum  $\mathbf{x}^*$ ;
2 Output: Updated sub-population  $P$ , final iterations  $\mathcal{I}$ ;
3  $\mathcal{I} \leftarrow \mathcal{I}_{current}$ ;  $\mathbf{x}_i^* \leftarrow$  optimum of the  $P_i$ ;
4 while  $\mathcal{I} < \mathcal{I}_{current} + \mathcal{I}_{max}$  do
5   for  $k \leftarrow 1, \dots, \lambda$  do
6     Generate sample point  $\mathbf{x}_k$  using reproduction operators in CMA-ES;
7     Calculate  $f(\mathbf{x}_k)$  and  $\rho(\mathbf{x}_k)$ ;
8     if  $\rho(\mathbf{x}_k) > 0$  then
9        $c = c + 1$ ; // Calculate the number of points sampled in high-density
        areas
10    end
11    if  $f(\mathbf{x}_k) > f(\mathbf{x}_i^*)$  then
12       $\mathbf{x}_i^* \leftarrow \mathbf{x}_k$ ;
13    end
14  end
15  Non-dominated sorting on  $\lambda$  sample points;
16  if  $c > \varepsilon_{hd,i}$  then
17    Return  $\emptyset$ ; // Discard the population  $P_i$ 
18  end
19   $\mathcal{I} \leftarrow \mathcal{I} + 1$ ;
20  if  $\text{mod}(\mathcal{I}, \mathcal{I}_{max}) = 0$  then
21    if  $\sigma < \varepsilon_\sigma$  and  $\lceil f(\mathbf{x}_i^*) \rceil \neq \lceil f(\mathbf{x}^*) \rceil$  then
22      Return  $\emptyset$ ;
23    end
24  end
25  Update sub-population  $P$  using CMA-ES operators;
26  if  $f(\mathbf{x}_i^*) > f(\mathbf{x}^*)$  then
27     $\mathbf{x}^* \leftarrow \mathbf{x}_i^*$ ; // Update potential global optimum
28  end
29 end
30 Return  $P, \mathcal{I}$ ;

```

collectively form $\mathcal{T} = \mathcal{T}_1 \cup \dots \cup \mathcal{T}_n$, which is employed to construct density space according to Equations (2) and (3).

Algorithm 3 describes the pseudocode of the third stage of DAEA, i.e., density-assisted restart, and its details are described as follows.

First, DAEA randomly samples k candidate solutions $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ and then calculates the density of these candidate solutions based on the exploration trajectories $\mathcal{T}$ of all historically extinct sub-populations. Next, the solution $\bar{\mathbf{x}}$ with the lowest density is selected, marking an unexplored area. Finally, $\bar{\mathbf{x}}$ is employed as the mean point of new sub-population P , and P is restarted by sampling λ points based on the normal distribution $\mathcal{N}(\bar{\mathbf{x}}, \sigma^2 \mathbf{I})$, where $\sigma = 0.5$ is the initial step size, $\mathbf{I}$ is the identity matrix, and λ is the sub-population size.

Algorithm 3: Density-Assisted Restart**Input:** Exploration trajectories $\mathcal{T}$ **Output:** Restarted sub-population P

- 1 Randomly sample candidate solutions $\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$;
- 2 Calculate the density $\{\rho(\mathbf{x}_1), \dots, \rho(\mathbf{x}_k)\}$ based on $\mathcal{T}$;
- 3 $\bar{\mathbf{x}} = \arg \min\{\rho(\mathbf{x}_1), \dots, \rho(\mathbf{x}_k)\}$;
- 4 Restart sub-population P based on the mean point $\bar{\mathbf{x}}$;
- 5 **Return** P

3.3 Dynamic Respond Strategy

3.3.1 Prediction Based on LSTM. The first stage of DAEA, i.e., density-assisted initialization, heavily relies on a substantial number of sampling points to capture the raw fitness landscape of the problem. Although it demonstrated satisfactory performance on static problems [Xu et al., 2021c], the limited computational resources available between successive environmental changes restrict the acquisition of fitness landscape information by the initial population.

To keep the density-assisted initialization applicable to DMMOPs, we employ the LSTM network to discover potential temporal patterns in dynamic optimization, which could reduce the budget allocated for fitness evaluations required by this strategy. Specifically, DAEA adopts the same population $\mathcal{P}_1$ generated by density-assisted initialization strategy for each environmental context. At each timestep t , the fitness values of all sample points are recorded as $\mathcal{D}_t = [f(\mathbf{x}_1, t), \dots, f(\mathbf{x}_{N_t}, t)]$. Then, the list of fitness values $\mathcal{D}_t$ within τ successive environments is denoted as $\mathcal{D} = [\mathcal{D}_1^T, \dots, \mathcal{D}_\tau^T]$. We defined a network $\mathcal{N}$ with the input layer, LSTM layer, fully connected layer, and regression layer. These fitness values are normalized and utilized as input data for the LSTM network $\mathcal{N}$.

For the prediction step τ_p , the fitness values $\mathcal{D}$ from the previous τ environments are used to train the network $\mathcal{N}$. Here, the fitness values matrix $[\mathcal{D}_1, \dots, \mathcal{D}_{\tau-\tau_p}]$ serves as inputs, while the fitness values $[\mathcal{D}_{\tau-\tau_p+1}, \dots, \mathcal{D}_\tau]$ from subsequent environments are used as outputs. Finally, the trained LSTM network $\mathcal{N}$ is used to predict the fitness values matrix $[\mathcal{D}_{\tau+1}, \dots, \mathcal{D}_{\tau+\tau_p}]$ for the population $\mathcal{P}_1$ over the upcoming τ_p environments.

After the prediction, it is necessary to evaluate whether the results are acceptable or not. We choose top s individuals as the validation set and calculate their descending rankings $\mathcal{R}_r = \{r_1, r_2, \dots, r_s\}$ and $\mathcal{R}_p = \{p_1, p_2, \dots, p_s\}$ by the real fitness value and the predicted fitness value, respectively. Then, the prediction accuracy a is calculated as follows:

$$a = 1 - \frac{\sum_{i=1}^s |r_i - p_i|}{s^2/2}. \quad (11)$$

If the accuracy a meets the threshold requirements, then this prediction can be considered acceptable, and the predicted fitness value is used without evaluating the remaining individuals. Otherwise, evaluate the remaining individuals.

3.3.2 Memory Strategy. The memory strategy is adopted to address the dynamic optimization with non-dramatically changing environments.

For the t th environment, the high-quality solutions in the previous environment are kept in the memory set $\mathcal{M}_t = \Phi_{t-1}$. Whenever the environment changes, after re-initializing the population $\mathcal{P}$, each sample point in the memory set $\mathcal{M}_t$ serves as a mean point $\bar{\mathbf{x}}_i$ to re-generate the new sub-population P_i to track the high-quality solution in the normal non-dramatic change mode.

3.4 The Framework of DAEA

DAEA is a typical niching-based multimodal optimization algorithm. The pseudocode of the complete framework of DAEA is shown in Algorithm 4, and the details are described as follows:

First (from lines 1 to 3), DAEA sets the environment counter $t = 1$, the algorithmic parameters, and LSTM network $\mathcal{N}$. Moreover, DAEA employs IDBPI to generate a set of uniform sampling points $\mathcal{P}_1$.

In the stage of density-assisted initialization (from lines 5 to 24), DAEA evaluates the population $\mathcal{P}_1$ directly in the previous τ environments and predicts the fitness value of population $\mathcal{P}_1$ using LSTM network $\mathcal{N}$ after the τ -th environment, where the network $\mathcal{N}$ is trained in the τ -th environment. Then, FDBPI is employed to generate another set of sampling points $\mathcal{P}_2$, which is merged with $\mathcal{P}_1$ to obtain the knowledge of the raw fitness landscape at the beginning of environments. Next, DAEA uses NBC [Preuss, 2010] to divide the population $\mathcal{P}$ into n sub-populations $\{P_1, \dots, P_n\}$. Moreover, DAEA generates $|\mathcal{M}_t|$ new sub-populations using memory set $\mathcal{M}_t$ in non-initial environments, where $|\mathcal{M}_t|$ represents the number of solutions in the memory set.

In the stage of density-assisted search (from lines 25 to 32), DAEA employs the multi-species cooperative coevolution [Xu et al., 2021c] to allocate computational resources to each sub-population. In each cycle, the best sub-population, i.e., the sub-population which holds the best solution of the population $\mathcal{P}$, and the most promising sub-population, i.e., the sub-population that converges to a global optimum with the least number of expected iterations, are allowed to iterate $\mathcal{I}_{\max}$. Then, the density-assisted search (Algorithm 2) is used to optimize two selected sub-populations. The best sub-population is removed from the sub-population set at convergence, and its best solution is preserved in the set Φ_t . In this article, a sub-population is considered to converge if the maximum difference between the fitness values in the sub-population is less than 10^{-6} .

In the stage of density-assisted restart (from lines 33 to 35), DAEA invokes the density-assisted restart (Algorithm 3) to regenerate a new sub-population when the sub-population set is empty and then goes to the stage of density-assisted search.

Finally, after the environment changes, the environment counter t and the memory set $\mathcal{M}$ are updated. When the termination conditions are satisfied, and the found high-quality solutions $\{\Phi_1, \Phi_2, \dots\}$ are returned.

3.5 Time Complexity of DAEA

The time complexity of the DAEA framework is influenced by five main components: density-assisted initialization, NBC, density-assisted search, density-assisted restart, and prediction mechanism. Here, we analyze the time complexity of each component.

- The density-assisted initialization consists of IDPIS and FDPIS. For IDPIS, the time complexity is determined by the necessity to sample k points and calculate their density for each new individual. Let N_1 represent the population size generated by IDPIS. Therefore, the time complexity of IDPIS is $O(N_1(kN_1))$. For FDPIS, the time complexity arises from calculating the density of m sampling points and the current population, and selecting N_2 individuals from them. This process occurs at the beginning of T environments. Consequently, the time complexity is $O(T(mN_1 + m \log(N_2)))$.
- NBC occurs at the beginning of T environments, so the time complexity is $O(T(N_1 + N_2)^2)$.
- The time complexity of the density-assisted search can be approximated as $O(T\bar{\mathcal{I}}n \cdot (\lambda n))$, where $\bar{\mathcal{I}}$ represents the mean number of iterations for each sub-population, λ denotes the sub-population size, n indicates the number of sub-populations, and $O(\lambda n)$ represents the time complexity of calculating densities of sample points.

Algorithm 4: The Framework of DAEA

```

1   $t \leftarrow 1$ ;
2  Initialize parameters and LSTM network  $\mathcal{N}$ ;
3  Generate initialize population  $\mathcal{P}_1$  using IDBPI;
4  while termination conditions are not satisfied do
5      if  $t \leq \tau$  then
6          Evaluate the population  $\mathcal{P}_1$ ;
7          if  $t = \tau$  then
8              Train LSTM network  $\mathcal{N}$ ;
9          end
10     else
11         Predict the fitness value of  $\mathcal{P}_1$  using  $\mathcal{N}$ ;
12         Calculate the accuracy of the prediction  $a$ ;
13         if  $a > \varepsilon_a$  then
14             Accept the prediction results;
15         else
16             Evaluate the remaining individuals;
17         end
18     end
19     Generate initialize population  $\mathcal{P}_2$  using FDBPI;
20      $\mathcal{P} \leftarrow \mathcal{P}_1 \cup \mathcal{P}_2$ ;
21      $\{P_1, \dots, P_n\} \leftarrow \text{NBC}(\mathcal{P})$ ; // Divide the population  $\mathcal{P}$  into  $n$  sub-populations
22     if  $t \geq 2$  then
23         Generate  $|\mathcal{M}_t|$  sub-populations using memory;
24     end
25     Optimize all sub-populations (Alg. 2); // Density-assisted search for  $n + |\mathcal{M}_t|$ 
        populations
26     while the environment is not changed do
27         Optimize the best sub-population (Alg. 2);
28         Optimize the most promising sub-population (Alg. 2);
29         if the best sub-population has converged then
30             Add the best solution to  $\Phi_t$ ;
31             Remove the best sub-population;
32         end
33         if the sub-population set is empty then
34             Restart a sub-population (Alg. 3);
35         end
36     end
37      $t \leftarrow t + 1$ ;
38     Update the memory set  $\mathcal{M}$ ;
39 end
40  $\Phi \leftarrow \{\Phi_1, \dots, \Phi_t\}$ 
41 Return  $\Phi$ 

```

- The time complexity of the density-assisted restart can be approximated as $O(T\bar{R}kn\bar{I})$, where $\bar{R}$ represents the mean number of restarts, $n\bar{I}$ is the size of the exploration trajectories $\mathcal{T}$ of all historically extinct sub-populations, and $O(kn\bar{I})$ represents the time complexity of calculating densities of sample points.
- The time complexity of LSTM prediction is estimated as $O(4\tau(h_D^2 + N_1h_D))$, where the input feature dimension is N_1A , the hidden state dimension is h_D , and τ is the sequence length.

3.6 Difference between DAEA and Other Dynamic Multimodal Optimization Algorithms

Several dynamic multimodal optimization algorithms have been proposed to improve computational efficiency and maintain population diversity [Cheng et al., 2018b; Cuevas et al., 2021; Luo et al., 2019]. However, our DAEA framework introduces several key innovations:

- Unlike existing techniques, which typically focus on specific components such as exclusion distances or penalty strategies, DAEA offers a complete framework encompassing population initialization, searching, and restarting. The density model plays a central role in maintaining population diversity at all evolutionary stages. First, existing techniques, such as Ahrari et al. [2017, 2022a, b], define exclusion regions for sub-populations to prevent overlap, or Wei et al. [2022], use penalties to guide populations away from elite-solution neighborhoods. However, these methods do not utilize the distribution of simultaneously active sub-populations. Second, DAEA jointly constructs a density space from all sub-populations except the current one and leverages density and fitness values to guide exploration into low-density regions, enhancing global diversity while avoiding redundant sampling. Third, for restarting, traditional strategies like random or heuristic restarts [Kazemi Kordestani et al., 2020; Yazdani et al., 2020] often lead to significant re-exploration of the solution space. In contrast, DAEA uses density landscapes from extinct sub-populations to reinitialize in low-density regions, significantly reducing the redundant computation and improving the capacity of discovering potential solutions. DAEA increases the possibility of jumping off the plateau by constructing density landscapes that ensure that populations preferentially explore the low-density areas of the plateau.
- While common dynamic response strategies rely on explicit archives [Cao et al., 2018; Vafashoar and Meybodi, 2020], computational resource reallocation [Yazdani et al., 2023b], or prediction [Liu et al., 2018; Liu et al., 2020; Meier and Kramer, 2020], DAEA introduces a novel approach. First, existing methods often focus on predicting solution position or environmental patterns, such as Ahrari et al. [2021] using multilevel prediction or Wu et al. [2023] employing kriging models and support vector machines for prediction. Moreover, DAEA instead predicts changes in fitness landscapes for initial populations across different environments, enabling efficient adaptation to dynamic conditions. Furthermore, unlike traditional memory strategies that directly reuse archived solutions, DAEA regenerates populations around archived solutions, effectively enhancing diversity while leveraging historical information.

4 Experiments

In this section, we first introduce the experimental design, including the benchmark suite, the evaluation metrics, the compared algorithms, and the parameter settings. Second, the experimental results are shown and analyzed.

4.1 Experimental Design

Benchmark. The benchmark suite of CEC'2022 competition on seeking multiple optima in dynamic environments is employed to test the performance of dynamic multimodal optimization

Table 1. Details of the Benchmark Suite of CEC'2022 Competition on Seeking Multiple Optima in Dynamic Environments

Group	Index	Multimodal Function	Dynamic Mode	Dimension
G1	P1	F1	Small step changes	5
	P2	F2	Small step changes	5
	P3	F3	Small step changes	5
	P4	F4	Small step changes	5
	P5	F5	Small step changes	5
	P6	F6	Small step changes	5
	P7	F7	Small step changes	5
	P8	F8	Small step changes	5
G2	P9	F8	Small step changes	5
	P10	F8	Large step changes	5
	P11	F8	Random changes	5
	P12	F8	Chaotic changes	5
	P13	F8	Recurrent changes	5
	P14	F8	Recurrent with noisy changes	5
	P15	F8	The number of global optima changes linearly	5
	P16	F8	The number of global optima changes randomly	5
G3	P17	F1	Small step changes	10
	P18	F2	Small step changes	10
	P19	F3	Small step changes	10
	P20	F4	Small step changes	10
	P21	F5	Small step changes	10
	P22	F6	Small step changes	10
	P23	F7	Small step changes	10
	P24	F8	Small step changes	10

algorithms. This benchmark suite consists of 24 DMMOPs, which include 8 dynamic modes and 8 multimodal functions. Specifically, the dynamic modes range from small to large step changes, from random, chaotic changes to recurrent or noisy recurrent changes, and from linear to stochastic adjustments that affect the globally optimal quantities. The multimodal functions are as follows: F1 to F4 are simple multimodal functions generated using the DF generator [Morrison and De Jong, 1999]. These functions have different numbers of global and local peaks, with peak heights, widths, and positions set according to specific requirements. F5 to F8 are complex multimodal functions, derived from a combination of functions in the CEC'2013 competition [Li et al., 2013]. These functions have a larger number of global peaks, and their fitness landscapes are adjusted by parameters such as λ and σ to increase the difficulty of finding the high-quality solution.

This benchmark suite can be classified into three groups (Table 1). The first group includes 8 different multimodal functions and the same dynamic mode. The second group includes 8 dynamic modes and the same multimodal function. The dimension of these two groups $D = 5$. The third group is similar to the first group, except that the dimension $D = 10$. Note that the environment changes when the number of fitness value evaluations in the current environment is exhausted. These groups are, respectively, denoted as G1, G2, and G3 in this section.

Evaluation Metrics. CEC'2022 dynamic multimodal optimization benchmark suite adopts the **Peak Ratio (PR)** as the performance metric, which is as follows [Luo et al., 2022a]:

$$PR_{\epsilon} = \frac{\sum_{i=1}^{run} \sum_{j=1}^{env} NPF_{ij}(\epsilon)}{\sum_{i=1}^{run} \sum_{j=1}^{env} PEAKS_{ij}}, \quad (12)$$

Table 2. Parameter Setting of DAEA on CEC'2022 Dynamic Multimodal Optimization Benchmark

Parameters	Meaning	Values
N_1	The size of population $\mathcal{P}_1$	0.4MaxFES
N_2	The size of population $\mathcal{P}_2$	0.1MaxFES
λ	The size of sub-population size	$7 + \lfloor (3 \log(D)) \rfloor$
$\mathcal{I}_{\max}$	The maximum number of iterations per cycle	20
k	The number of sampling	10 FES _{rest}
h	The bandwidth	1.1
α	The scale factor	100
ε_σ	The evolutionary step threshold	0.02
ε_a	The accuracy threshold	0.5
τ_p	The prediction step	10
s	The validation set size	2,000

where $run = 30$ represents the number of requested independent runs of the algorithm on a DMMOP, $env = 60$ represents the total number of environments of a DMMOP, $NPF_{ij}(\varepsilon)$ represents the number of found optimal solutions in the i th run and the j th environment at the accuracy level ε , and $PEAKs_{ij}$ represents the total number of global peaks in the i th run and the j th environment.

Moreover, the maximum number of fitness evaluations for each environment is set to MaxFES = 5,000 D , and the random seeds are set from 1 to 30. It is worth noting that the benchmark set is aligned with the CEC'2022 dynamic multimodal optimization competition [Luo et al., 2022a].

Compared Algorithms. The participating algorithms of CEC'2022 and CEC'2023 competitions on seeking multiple optima in dynamic environments are selected as compared algorithms,¹ including the winner AMLP-RS-CMSA-ESII [Ahraria et al., 2022a, b], runner-up AMPSO, and third place winners NBNC-EA-SVR of the CEC'2022 competition, and the winner CMA-ES-DS and runner-up mGA of the CEC'2023 competition. Moreover, other participating algorithms, such as mEGA, DMMDE [Lin et al., 2022b], and HmSO, as well as a state-of-the-art dynamic multimodal optimization algorithm KMEA-SVM [Wu et al., 2023] are also included in the comparison experiments.

Parameter Setting. The parameters in this experiment are set as follows and shown in Table 2. The size of population $\mathcal{P}_1$ is set to 0.4MaxFES, the size of population $\mathcal{P}_2$ is set to 0.1MaxFES, the size of sub-population $\lambda = 7 + \lfloor (3 \log(D)) \rfloor$, the maximum number of iterations per cycle $\mathcal{I}_{\max} = 20$, the number of sampling $k = 10\text{FES}_{\text{rest}}$, where FES_{rest} means the rest number of the fitness evaluations for each environment, the bandwidth h is set to 1.1, the scale factor $\alpha = 100$, the evolutionary step threshold $\varepsilon_\sigma = 0.02$, the accuracy threshold ε_a is set to 0.5, the τ_p is set to 10, the validation set size s is set to 2,000, the input and output layers of the neural network $\mathcal{N}$ are set to 40 and 10, respectively, and the hidden layer is set to 300.

4.2 Experimental Results

Table 3 presents the experimental results of the proposed DAEA on the CEC'2022 dynamic multimodal optimization benchmark at three accuracy levels $\varepsilon = \{10^{-3}, 10^{-4}, 10^{-5}\}$. The results encompass the median, mean, best, and worst found PR over 30 runs, where the values equal to 1 are highlighted in bold. In general, the experimental results indicate that DAEA achieves consistent

¹The results of the CEC'2022 and CEC'2023 dynamic multimodal optimization competitions are available at <https://github.com/MiLabHITSZ/dmmops>.

Table 3. The Median, Mean, Best, and Worst Found Peak Ratio of DAEA on CEC'2022 Dynamic Multimodal Optimization Benchmark at Three Different Accuracy Levels $\varepsilon = \{10^{-3}, 10^{-4}, 10^{-5}\}$

Groups	Problem	10^{-3}				10^{-4}				10^{-5}			
		Median	Mean	Best	Worst	Median	Mean	Best	Worst	Median	Mean	Best	Worst
G1	P1	0.750	0.753	0.788	0.733	0.753	0.753	0.788	0.733	0.753	0.753	0.788	0.733
	P2	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	P3	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	P4	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
	P5	0.656	0.653	0.667	0.619	0.653	0.653	0.667	0.619	0.653	0.653	0.667	0.619
	P6	0.339	0.337	0.377	0.285	0.337	0.337	0.377	0.285	0.337	0.337	0.377	0.285
	P7	0.667	0.665	0.667	0.658	0.665	0.665	0.667	0.658	0.665	0.665	0.667	0.658
	P8	0.685	0.687	0.710	0.656	0.687	0.687	0.710	0.656	0.687	0.687	0.710	0.656
G2	P9	0.685	0.687	0.710	0.656	0.687	0.687	0.710	0.656	0.687	0.687	0.710	0.656
	P10	0.638	0.636	0.665	0.615	0.636	0.636	0.665	0.615	0.636	0.636	0.665	0.615
	P11	0.607	0.605	0.640	0.571	0.605	0.605	0.640	0.571	0.605	0.605	0.640	0.571
	P12	0.584	0.584	0.623	0.556	0.584	0.584	0.623	0.556	0.584	0.584	0.623	0.556
	P13	0.645	0.645	0.677	0.617	0.645	0.645	0.677	0.617	0.645	0.645	0.677	0.617
	P14	0.623	0.622	0.654	0.577	0.622	0.622	0.654	0.577	0.622	0.622	0.654	0.577
	P15	0.480	0.480	0.507	0.443	0.480	0.480	0.507	0.443	0.480	0.480	0.507	0.443
	P16	0.586	0.584	0.609	0.558	0.584	0.584	0.609	0.558	0.584	0.584	0.609	0.558
G3	P17	0.663	0.649	0.767	0.492	0.649	0.649	0.767	0.492	0.649	0.649	0.767	0.492
	P18	0.975	0.971	0.992	0.950	0.971	0.971	0.992	0.950	0.971	0.971	0.992	0.950
	P19	0.842	0.838	0.879	0.783	0.838	0.838	0.879	0.783	0.838	0.838	0.879	0.783
	P20	0.954	0.951	0.967	0.879	0.951	0.951	0.967	0.879	0.951	0.951	0.967	0.879
	P21	0.553	0.565	0.642	0.464	0.565	0.565	0.642	0.464	0.565	0.565	0.642	0.464
	P22	0.427	0.428	0.454	0.406	0.428	0.428	0.454	0.406	0.428	0.428	0.454	0.406
	P23	0.600	0.602	0.642	0.575	0.602	0.602	0.642	0.575	0.602	0.602	0.642	0.575
	P24	0.517	0.524	0.608	0.490	0.524	0.524	0.608	0.490	0.524	0.524	0.608	0.490

results at different accuracy levels. This reliability stems from DAEA preserving the high-quality solution after reaching high convergence.

DAEA achieves PR = 1 on problems P2, P3, and P4, indicating its superior exploration ability in addressing low-dimensional simple problems. Importantly, this effectiveness extends to corresponding high-dimensional problems, with PRs of 0.971, 0.838, and 0.951 on problems P18, P19, and P20, respectively.

For complex composition problems (P5–P16, P21–P24), the majority of DAEA results fall within the range [0.5, 0.6], signifying the discovery of more than half of the global optimal peaks, except for unexpectedly lower performances on P6, P15, and P22. Furthermore, the analysis of the best and worst PR reveals that DAEA exhibits a relatively stable performance on low-dimensional problems, with fluctuations below 0.1. However, as dimensionality increases, the stability of DAEA tends to diminish.

To underscore the superiority of DAEA, Table 4 lists the PRs of DAEA and the algorithms that are mentioned in Section 4.1 Experimental Design—Compared Algorithms at the commonly used accuracy level $\varepsilon = 10^{-4}$. Notably, the best results in each row of all the tables are in bold for emphasis. The penultimate row tabulates the frequency with which each algorithm achieves the **best PRs (bprs)** across the 24 DMMOPs. The final row provides the overall rankings of all algorithms determined through non-parametric statistical hypothesis testing, i.e., the Friedman test.

In the first group of DMMOPs, DAEA secures the top position across six problems, i.e., P2–P5, P7, and P8, except for being surpassed by CMA-ES-DS and AMPSO on problem P1, and by CMA-ES-DS and AMLP-RS-CMSA-ESII on problem P6. In the second group of DMMOPs, DAEA demonstrates superior performance, outperforming other compared algorithms on all eight problems. These

Table 4. The PRs of DAEA and All Compared Algorithms on the CEC'2022 Dynamic Multimodal Optimization Benchmark at Accuracy Level $\varepsilon = 10^{-4}$

Groups	Problem	DAEA	CMA-ES-DS ^a	mGA ^a	AMLPS-ESII ^a	AMPSO ^a	mEGA ^a	NBNC-EA-SVR ^a	DMMDE ^a	HmSO ^a	KMEA-SVM ^b
G1	P1	0.753	0.767	0.613	0.696	0.765	0.560	0.543	0.403	0.622	0.585
	P2	1.000	1.000	0.867	0.997	0.999	0.690	0.507	0.807	0.973	0.521
	P3	1.000	1.000	0.842	0.954	0.999	0.739	0.521	0.449	0.973	0.558
	P4	1.000	1.000	0.873	0.970	0.999	0.752	0.526	0.488	0.987	0.546
	P5	0.653	0.617	0.321	0.547	0.546	0.297	0.582	0.241	0.564	0.588
	P6	0.337	0.402	0.243	0.461	0.263	0.240	0.271	0.249	0.254	0.271
	P7	0.665	0.594	0.312	0.634	0.606	0.261	0.549	0.279	0.565	0.566
	P8	0.687	0.629	0.033	0.608	0.433	0.030	0.357	0.183	0.223	0.365
G2	P9	0.687	0.629	0.034	0.608	0.433	0.030	0.357	0.183	0.223	0.364
	P10	0.636	0.563	0.006	0.600	0.274	0.005	0.320	0.191	0.081	0.322
	P11	0.605	0.554	0.008	0.564	0.247	0.007	0.306	0.179	0.074	0.308
	P12	0.584	0.579	0.006	0.563	0.267	0.004	0.320	0.198	0.053	0.321
	P13	0.645	0.621	0.012	0.570	0.297	0.010	0.326	0.181	0.110	0.331
	P14	0.622	0.619	0.017	0.580	0.275	0.014	0.324	0.183	0.096	0.329
	P15	0.480	0.377	0.040	0.365	0.178	0.035	0.223	0.111	0.087	0.225
	P16	0.584	0.549	0.039	0.533	0.205	0.034	0.323	0.177	0.118	0.325
G3	P17	0.649	0.696	0.373	0.505	0.542	0.167	0.334	0.250	0.143	0.334
	P18	0.971	0.600	0.661	0.938	0.733	0.120	0.516	0.478	0.075	0.520
	P19	0.838	0.817	0.688	0.873	0.712	0.299	0.540	0.331	0.223	0.558
	P20	0.951	0.975	0.815	0.950	0.770	0.298	0.515	0.330	0.217	0.525
	P21	0.565	0.597	0.330	0.420	0.335	0.327	0.325	0.324	0.233	0.326
	P22	0.428	0.363	0.249	0.581	0.250	0.245	0.269	0.230	0.147	0.270
	P23	0.602	0.597	0.372	0.651	0.534	0.326	0.468	0.167	0.117	0.469
	P24	0.524	0.404	0.039	0.527	0.344	0.028	0.371	0.150	0.064	0.381
bprs		15	7	0	5	0	0	0	0	0	0
Friedman		1.625	2.167	7.333	2.792	4.667	9.042	6.292	8.125	7.667	5.292

^a The top-ranked algorithms participating in the CEC'2022 and CEC'2023 competition on seeking multiple optima in dynamic environments, and the results are available at <https://github.com/MiLabHITSZ/dmmops>.

^b The result of KMEA-SVM in the paper [Wu et al., 2023].

results suggest the effectiveness of DAEA in handling diverse dynamic modes within DMMOPs. In the third group of DMMOPs, DAEA achieved optimal results solely on P18, securing the second position on the remaining problems. Specifically, it ranked below the CEC'2023 dynamic multimodal optimization competition winner CMA-ES-DS on P17, P20, and P21, and the CEC'2022 winner AMLP-RS-CMSA-ESII on P19 and P22–P24. Furthermore, the Friedman rank of DAEA suggests that the algorithm consistently achieves stable and superior performance rankings on this benchmark suite.

A pairwise comparison of DAEA against the winners of the CEC'2022 and CEC'2023 dynamic multimodal optimization competitions, i.e., AMLP-RS-CMSA-ESII and CMA-ES-DS, clearly reveals the superiority of DAEA. In both comparisons, DAEA outperforms them on 19 problems and falls short on 5 problems in both comparisons. Compared to the other algorithms, DAEA outperforms on all functions except P1 where it is inferior to AMPSO. Additionally, Table 5 presents the Wilcoxon signed-rank test for the pairwise comparison of DAEA with all other algorithms, highlighting results where DAEA outperforms significantly at a 0.05 significance level. The Holm-Bonferroni correction method has been employed to control for the increase in Type I error during hypothesis testing. The results suggest that DAEA shows significant advantages over other algorithms at the 0.05 significance level, and its superior performance is robustly validated.

Table 5. Results Obtained by the Wilcoxon Signed-Rank Test for the Pairwise Comparisons between DAEA and Compared Algorithms

vs.	R^+	R^-	p-Value	Holm
mGA	300.0	0.0	0.000	0.000
AMPSO	296.0	4.0	0.000	0.000
mEGA	300.0	0.0	0.000	0.000
NBNC-EA-SVR	300.0	0.0	0.000	0.000
DMMDE	300.0	0.0	0.000	0.000
HmSO	300.0	0.0	0.000	0.000
KMEA-SVM	300.0	0.0	0.000	0.000
CMA-ES-DS	219.5	56.5	0.011	0.022
AMLPS-RS-CMSA-ESII	231.5	68.5	0.018	0.018

5 Discussion

In this section, we first conduct an ablation study to analyze the three density-assisted phases of DAEA. Next, we examine the role of the dynamic response strategy employed within the algorithm. Finally, we perform a parameter analysis to explore the impact of different parameters on the performance of DAEA.

5.1 Ablation Study

DAEA comprises three integral stages, i.e., density-assisted initialization, search, and restart. In this section, we conduct an ablation study to scrutinize the influence of these three stages on the performance of DAEA. We introduce three compared algorithms, denoted as DAEA-1, DAEA-2, and DAEA-3. Specifically, DAEA-1 substitutes the density-assisted search strategy with the original search strategy, DAEA-2 replaces the density-assisted initialization with a pseudo-random number generator, and DAEA-3 omits the density-assisted restart. To ensure equitable comparisons, all comparison algorithms abstain from using a prediction strategy and employ identical parameters as outlined in Section 4.

Figure 4 displays the experimental results for the four comparison algorithms. Comparison between DAEA and DAEA-1 evidences that a notable decline in performance occurs for all problems when the density-assisted search is omitted. This observation demonstrates that density-assisted search contributes significantly to enhancing the exploration efficiency of DAEA, which is attributed to its ability to improve the utilization of computational resources and promote the global diversity of candidate solutions.

Comparisons between DAEA and DAEA-3 reveal that, in numerous problems (e.g., P6, P8–P16, and P21–P24), DAEA-3 displays varying degrees of performance degradation compared to DAEA. The experimental results suggest that, although the impact of density-assisted restart on performance improvement is not as pronounced as density-assisted search, it still exerts a positive effect. Notably, density-assisted restart, recommended for application in complex problems to mitigate challenges arising from the loss of global diversity, contributes to a performance gain across nearly all composition problems, excluding only P5 and P7.

The density-assisted initialization, i.e., DAEA-2, demonstrates the performance improvement compared to DAEA specifically on P11–P14. However, the impact on performance in the context of other problems is not immediately evident. This observation implies that the uniform initial population generated by the density-assisted initialization may have favorable implications for

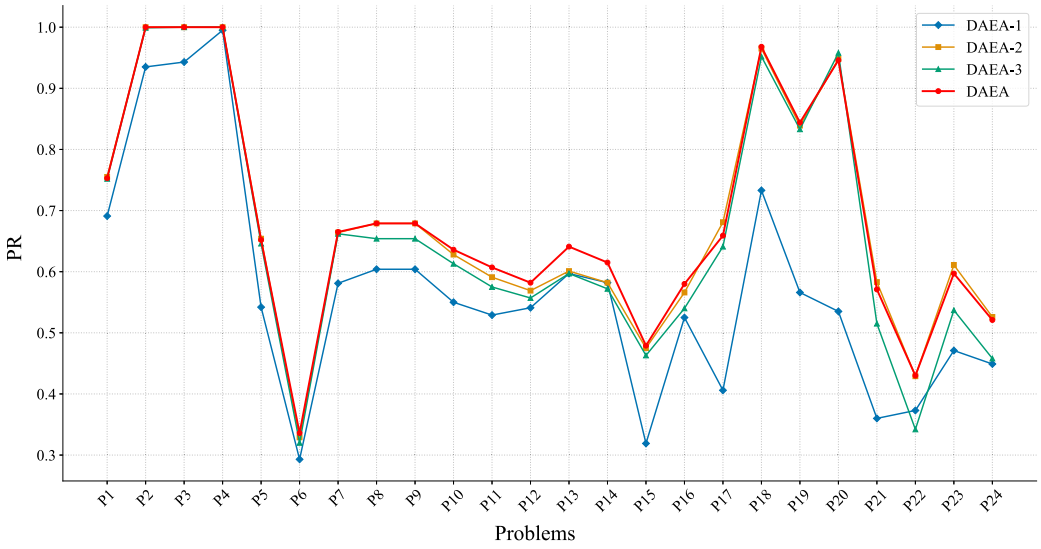


Fig. 4. The PRs of DAEA and ablation algorithms on the CEC'2022 dynamic multimodal optimization benchmark at accuracy level $\varepsilon = 10^{-4}$, where DAEA-1, DAEA-2, and DAEA-3 represent DAEA omitted the density-assisted search, initialization, and restart, respectively.

Table 6. Results Obtained by the Wilcoxon Signed-Rank Test for the Pairwise Comparisons between DAEA and Ablation Algorithms

vs.	R^+	R^-	p-Value
DAEA-1	300.0	0.0	0.000
DAEA-2	179.0	97.0	0.195
DAEA-3	290.5	9.5	0.000

randomly and periodically changing fitness landscapes when compared with initialization methods relying on pseudo-random number generators.

Furthermore, Table 6 provides the results of paired Wilcoxon signed-rank tests for DAEA and the three ablation algorithms, highlighting the superior results of DAEA at a significance level of 0.05. The statistical results reveal that DAEA outperforms DAEA-1 and DAEA-3 at the 0.05 significance level, underscoring the positive impact of the density-assisted search and restart methods proposed in this article. Notably, DAEA also outperforms DAEA-2, even though there is no significance level, i.e., $p > 0.05$, in the pairwise comparison between DAEA and DAEA-2, as indicated by the R^+ value. Overall, the experimental results, both in Figure 4 and in Table 6, support the favorable efficacy of the proposed density-assisted strategy for the EAs.

5.2 Analysis of Dynamic Response Strategy

DAEA employs two dynamic response strategies, i.e., LSTM-based prediction strategy and memory strategy. In this section, we discuss the effects of these two strategies on DAEA.

Table 7 presents the results of the comparison between DAEA and DAEA without prediction. Note that DAEA incorporates prediction solely in the last τ_p environments. Thus, the reported

Table 7. The Mean PRs of DAEA and DAEA without Prediction on the CEC'2022 Dynamic Multimodal Optimization Benchmark at Accuracy Level $\varepsilon = 10^{-4}$ in the Last τ_p Environment

Groups	Problem	DAEA	DAEA without Prediction
G1	P1	0.750	0.750
	P2	1.000	1.000
	P3	1.000	1.000
	P4	1.000	1.000
	P5	0.639	0.633
	P6	0.365	0.353
	P7	0.666	0.664
	P8	0.713	0.682
G2	P9	0.713	0.682
	P10	0.638	0.637
	P11	0.603	0.602
	P12	0.551	0.547
	P13	0.671	0.645
	P14	0.648	0.616
	P15	0.464	0.452
	P16	0.467	0.483
G3	P17	0.844	0.816
	P18	0.865	0.833
	P19	0.750	0.750
	P20	0.762	0.763
	P21	0.641	0.644
	P22	0.407	0.391
	P23	0.594	0.586
	P24	0.521	0.513
bprs		21	8

results only pertain to these last τ_p environments, as both algorithms yield identical results for the preceding unpredicted environments. The other experimental setup adheres to the specifications outlined in Section 4.

As depicted in Table 7, DAEA achieves the best PRs on 21 problems and demonstrates performance improvements on 16 problems, with notable enhancements observed on P6, P8, P9, P13–P15, P18, P19, and P22. These experimental results suggest the capability of prediction strategy to recognize and adeptly address various dynamic modes and complex fitness landscapes.

Given that the second group of DMMOPs (P9–P16) shares the same multimodal optimization function but incorporates eight dynamic change modes, we utilize this set to conduct a more in-depth analysis of the LSTM-based prediction strategy. As outlined in the benchmark suite description [Luo et al., 2022a], this group can be categorized into four sub-groups. P9 and P10 represent small and large step changes, P11 and P12 encompass random and chaotic changes, P13 and P14 involve recurrent without and with noise changes, and P15 and P16 entail linear and random changes in the number of global optima, respectively. For these problems, DAEA exhibits greater improvement on P9, P13–P15, lesser enhancement on P10–P12, and experiences a performance decline on P16. Note that both the first and third groups of DMMOPs use small step change mode. The experimental results indicate that the prediction strategy employed by DAEA effectively captures small step and periodic dynamic patterns, yet struggles to adapt to large step

Table 8. The Mean PRs of DAEA and DAEA without Memory on the CEC'2022 Dynamic Multimodal Optimization Benchmark at Accuracy Level $\varepsilon = 10^{-4}$

Groups	Problem	DAEA	DAEA without Memory
G1	P1	0.753	0.697
	P2	1.000	1.000
	P3	1.000	0.995
	P4	1.000	0.996
	P5	0.653	0.574
	P6	0.337	0.304
	P7	0.665	0.607
	P8	0.687	0.643
G2	P9	0.687	0.643
	P10	0.636	0.608
	P11	0.605	0.589
	P12	0.584	0.585
	P13	0.645	0.628
	P14	0.622	0.619
	P15	0.480	0.334
	P16	0.584	0.556
G3	P17	0.649	0.417
	P18	0.971	0.894
	P19	0.838	0.606
	P20	0.951	0.566
	P21	0.565	0.371
	P22	0.428	0.421
	P23	0.602	0.525
	P24	0.524	0.478
bprs		23	2

changes and stochastic or chaotic changes. This observation aligns with the inherent properties of LSTM.

Despite LSTM's inapplicability to certain change patterns, its selection as a prediction strategy in DAEA is justified for several reasons. First, DAEA relies on the relative ordering rather than the exact fitness values of initial populations for population clustering, enabling effective characterization of peak positions. Second, DAEA employs accuracy as a criterion to evaluate the acceptability of predictions. Last, while LSTM prediction may pose a risk in scenarios where it is not applicable, the saved fitness evaluations for restarting ensure the overall robustness of the algorithm.

Table 8 illustrates the influence of the memory strategy on DAEA. The compared algorithm employs DAEA without memory strategy, while the remaining experimental setup aligns with the specifications outlined in Section 4.

The experimental results presented in Table 8 indicate that DAEA performs worse than DAEA without memory only on P12, but it displays a significant improvement for all other problems. These results suggest the effectiveness of the memory strategy in swiftly identifying and tracking changing global high-quality solutions.

This effect is particularly pronounced in high-dimensional problems with small step change patterns, such as P20 and P21. The reason lies in the challenges posed by limited computational resources and complex fitness landscapes for the initial populations in locating peaks. However, in scenarios with small step change patterns, where the optimal solution undergoes small shifts,

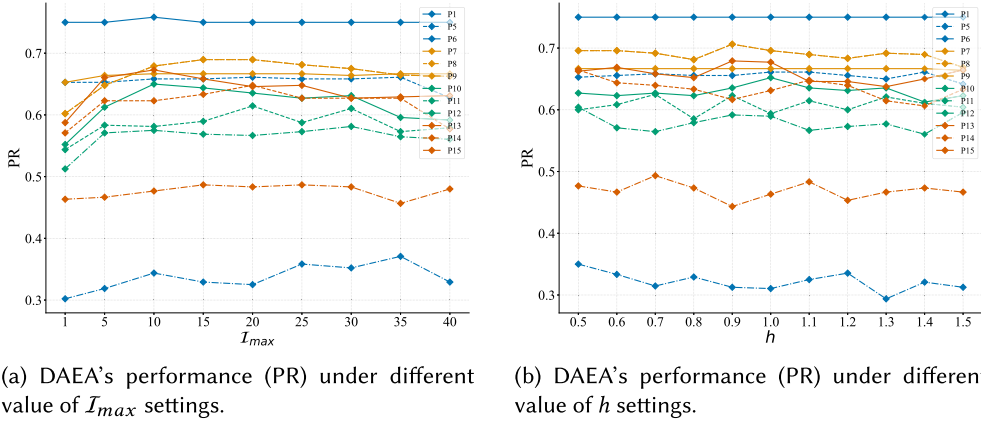


Fig. 5. Impact of important parameters of DAEA framework on performance with different value settings.

regenerating the population at historical locations compared to aimless search in the global space enhances the possibility of capturing the peak.

5.3 Parameter Analysis

In this section, we discuss the parameter settings for DAEA to provide guidance for practical applications. Some components of DAEA, such as the IDPIS, CMA-ES, and LSTM, adopt parameters directly from the original literature. Other parameters, such as k , α , and ε_σ , are empirically determined and should be fine-tuned based on the desired solution accuracy for specific problems. This section focuses on analyzing two key parameters, i.e., the maximum iterations per cycle (I_{max}) and the bandwidth (h). The analysis examines their influence on the performance of the DAEA framework across 12 benchmark functions, specifically P_1 and P_5 through P_{15} . The parameter I_{max} is tested with values in $\{1, 5, 10, 15, 20, 25, 30, 35, 40\}$, while h is tested with values in $\{0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5\}$. The results are illustrated in Figure 5(a) and (b).

The results in Figure 5(a) indicate that, for many problems, as I_{max} increases, the PR initially increases and then decreases, suggesting that a moderate value for I_{max} is optimal, with the best performance observed within the range $[10, 30]$. When I_{max} is small, frequent switching between sub-population optimizations occurs, while a larger I_{max} may lead to excessive computational resources being wasted on sub-populations that are not performing well. The results in Figure 5(b) show that, for different problems, the PR value fluctuates as h increases, without exhibiting a significant trend, although the fluctuations are not drastic. In fact, h primarily affects the determination of high-density regions. In practical applications, the choice of h depends largely on the decision maker's requirement for solution diversity, i.e., the distance between solutions in the decision space.

6 Application

In this section, we apply the DAEA to address the BAP at a container terminal [Ji et al., 2017; Ji et al., 2022]. The BAP exhibits dynamic characteristics, as the set of vessels requiring scheduling continuously changes within the tidal window [Qu et al., 2023]. Additionally, the problem is multimodal, necessitating the generation of multiple candidate solutions to allow decision-makers to adapt to unexpected events, such as quay crane malfunctions, which demand immediate transitions to alternative plans.

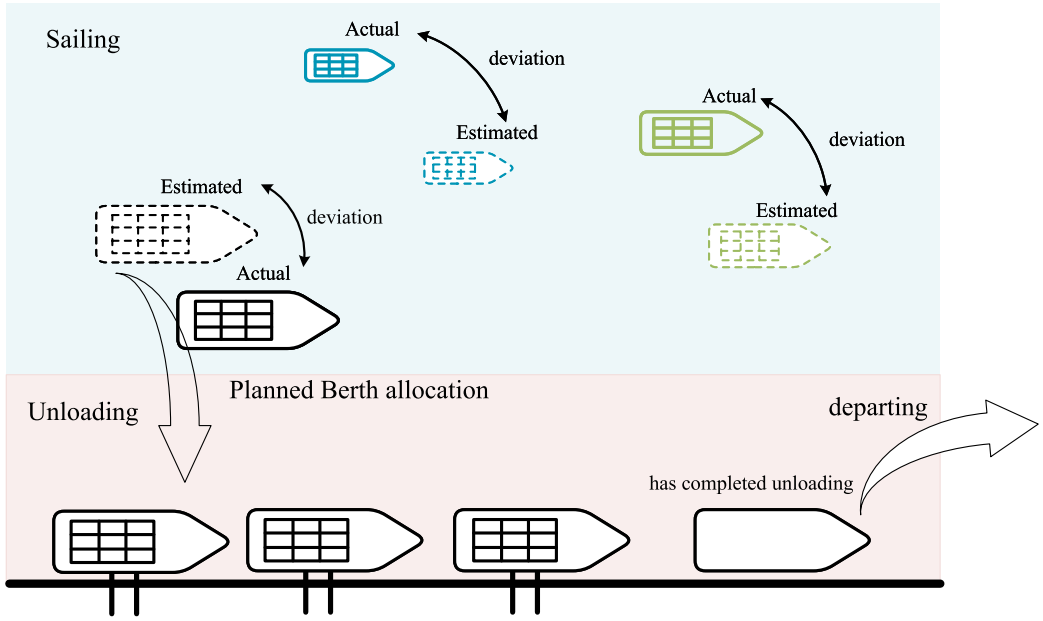


Fig. 6. Schematic representation of the BAP. The upper section illustrates the vessel's sailing status, highlighting the expected arrival (dotted line), the actual arrival (solid line), and the deviation between the two. The lower section depicts the vessel's unloading operations at the berth, including the planned berth allocation and the departure process following the completion of unloading.

6.1 Problem Description

The BAP is a critical task in container terminal operations, which involves allocating berths to arriving vessels to minimize their port stay duration and improve the terminal's operational efficiency. Figure 6 provides a schematic representation of this problem. Due to uncertainties in the maritime environment, actual arrival times and positions of vessels often deviate from their scheduled estimates, leading to timetable inconsistencies.

We model the BAP as a DMMOP, aiming to optimize berth allocation for vessels within tidal windows. Specifically, we consider vessels eligible for berthing during the next tidal peak, with the problem dynamically updated over time [Qu et al., 2023]. The following assumptions are made: the total number of tidal windows, T , is fixed, along with the set of vessels V_t and the number of vessels n_t arriving in each time window t . The coastline length L is constant. Additionally, no delay penalty is incurred if a vessel begins processing immediately upon arrival, resulting in zero waiting time under these conditions.

The mathematical model for minimizing the total waiting time of vessels in each tidal window t can be reformulated as follows:

$$\begin{aligned}
 & \min \sum_{i=1}^{n_t} s_{it} - a_{it}, \quad \forall t \in \{1, 2, \dots, T\}, \\
 & s.t. \begin{cases} s_{it} \geq a_{it}, & \forall i \in T_t, \forall t \in \{1, 2, \dots, T\}, \\ 0 \leq x_{it} \leq L - l_i, & \forall i \in T_t, \forall t \in \{1, 2, \dots, T\}, \\ x_{it} + l_i \leq x_{jt}, & \forall i, j \in T_t, i \neq j, \forall t \in \{1, 2, \dots, T\}, \\ s_{it} \geq 0, a_{it} \geq 0, & \forall i \in T_t, \forall t \in \{1, 2, \dots, T\}. \end{cases}
 \end{aligned}$$

where s_{it} represents the departure time, and a_{it} represents the arrival time of vessel i in time window t . The constraints ensure the feasibility of the vessel scheduling and spatial allocation at the terminal. Specifically, the start time s_{it} must not precede arrival a_{it} , berthing positions x_{it} are restricted by the coastline length L , while vessels i, j cannot overlap at the same berth, where l_i is the length of vessel i . All time-related variables must remain non-negative to ensure logical consistency.

6.2 Experimental Design and Performance Measurements

We selected NBSO2A [Jin et al., 2024], CSO [Cheng and Jin, 2015], RTPSO [Malekahmadi et al., 2020], and RCIPSO [Yang et al., 2024] as comparative algorithms to evaluate the performance and solution diversity of DAEA on the BAP [Zou et al., 2024]. Among these, NBSO2A is a dynamic multimodal optimization algorithm, RTPSO is specifically developed for the BAP, and CSO and RCIPSO are advanced variants of PSO.

For the experimental setup, we followed the generic settings outlined in the aforementioned algorithms. For algorithms with a fixed population size, we set the population size $n_{pop} = 100$ and the number of iterations to $\text{MaxIter} = 10D$, where D represents the total number of vessels processed in a time window. For algorithms with variable population size, we set the maximum number of fitness function evaluations to $\text{MaxFes} = 1,000D$. The total number of time windows T is set to 15. Note that the other parameters of DAEA are consistent with those in Section 4. Each algorithm was repeated for 30 independent runs on each BAP case.

Because of the complexity and uncertainty of the objective function in the real problem, the exact optimal solution cannot be determined. Consequently, we used two performance metrics to evaluate the algorithms: The first metric is the best fitness, which refers to the fitness value of the most adapted individual in the current population, assessing the algorithm's ability to converge to the high-quality solution. The second metric is the number of distinct peaks found and their mean fitness values, which gauges the multimodal algorithm's ability to preserve population diversity and explore the solution space effectively, thereby identifying multiple peaks.

6.3 Results

Table 9 presents the experimental results of the proposed DAEA and several compared algorithms on eight BAP cases. For each case, the set of global and local optima identified by comparison algorithms in an environment is denoted as Φ_t . The metrics reported include the average best fitness values from Φ_t across T environments over multiple runs, the overall mean fitness value, and the average size of Φ_t , represented as best fitness, mean fitness, and peaks found, respectively. Notably, only DAEA and NSO2A are designed for dynamic multimodal optimization, resulting in Peaks Found > 1 , while the other three algorithms yield Peaks Found = 1.

The results in Table 9 show that DAEA outperforms all other algorithms across all test cases. It identifies more global and local approximate optimal solutions and achieves higher quality solutions compared to RTPSO, RCIPSO, and CSO, which are designed solely for global optimization. This superior performance stems from DAEA's ability to adapt to the dynamic nature of the problem by capturing environmental change patterns, thereby enhancing optimization efficiency. For the "Peaks Found" metric, DAEA identifies approximately seven distinct solutions across different cases, demonstrating greater stability compared to CMA-ES-DS and NBSO2A. This stability arises because DAEA's discovery of a local optimum is closely tied to the convergence of a single sub-population. For similar problem types, the number of evaluations required for sub-population convergence does not vary significantly. Moreover, the quality of the approximate local optima identified by DAEA is superior, with the average fitness value remaining within twice the global optimum.

Table 9. The Best Fitness, Mean Fitness, and the Peaks Found of the DAEA, CMA-ES-DS, NBSO2A, RTPSO, RCIPSO, and CSO on F1-F8 BAP Cases

Problem	Measurement	DAEA	CMA-ES-DS	NBSO2A	RTPSO	RCIPSO	CSO
F1	Best Fitness	5.74E+03±1.57E+03	8.75E+03±1.92E+03	1.40E+04±1.98E+03	1.22E+04±1.43E+03	1.93E+04±4.22E+03	1.02E+04±1.62E+03
	Mean Fitness	8.16E+03±3.68E+03	9.28E+03±2.41E+03	4.78E+04±3.56E+04	-	-	-
	Peaks Found	7.41±0.55	6.93±0.81	3.27±2.00	-	-	-
F2	Best Fitness	4.23E+03±1.38E+03	5.36E+03±1.47E+03	1.04E+04±1.50E+03	9.37E+03±1.55E+03	1.57E+04±2.64E+03	7.27E+03±1.33E+03
	Mean Fitness	6.31E+03±2.72E+03	7.12E+03±2.93E+03	2.85E+04±2.00E+04	-	-	-
	Peaks Found	7.51±0.60	5.28±0.33	2.67±1.68	-	-	-
F3	Best Fitness	2.95E+03±1.37E+03	5.11E+03±1.64E+03	1.00E+04±1.91E+03	8.68E+03±1.78E+03	1.45E+04±2.57E+03	6.80E+03±1.56E+03
	Mean Fitness	5.14E+03±5.19E+03	9.28E+03±2.61E+03	4.33E+04±2.75E+04	-	-	-
	Peaks Found	7.61±0.63	3.17±1.19	4.27±2.85	-	-	-
F4	Best Fitness	5.74E+03±1.65E+03	7.95E+03±2.52E+03	1.42E+04±2.37E+03	1.33E+04±2.65E+03	2.07E+04±4.16E+03	1.01E+04±2.14E+03
	Mean Fitness	7.99E+03±3.85E+03	1.22E+04±7.40E+03	3.43E+04±2.30E+04	-	-	-
	Peaks Found	7.36±0.51	6.84±0.729	2.40±1.76	-	-	-
F5	Best Fitness	4.49E+03±1.38E+03	5.30E+03±1.40E+03	1.24E+04±1.68E+03	1.07E+04±1.65E+03	1.79E+04±2.62E+03	8.20E+03±9.41E+02
	Mean Fitness	6.28E+03±2.95E+03	8.67E+03±1.33E+03	3.95E+04±2.30E+04	-	-	-
	Peaks Found	7.37±0.49	6.13±0.90	3.00±1.57	-	-	-
F6	Best Fitness	1.85E+04±4.70E+03	2.12E+04±6.01E+03	3.76E+04±5.24E+03	3.56E+04±4.78E+03	5.19E+04±7.06E+03	3.01E+04±4.43E+03
	Mean Fitness	2.25E+04±1.12E+04	2.56E+04±0.94E+04	4.95E+04±2.17E+04	-	-	-
	Peaks Found	7.19±0.39	6.72±0.37	1.33±0.60	-	-	-
F7	Best Fitness	2.80E+04±6.23E+03	4.12E+04±8.53E+03	4.64E+04±4.75E+03	4.35E+04±6.44E+03	6.10E+04±7.52E+03	3.70E+04±4.72E+03
	Mean Fitness	3.26E+04±1.24E+04	3.76E+04±1.29E+04	6.20E+04±2.97E+04	-	-	-
	Peaks Found	7.17±0.42	5.85±0.94	1.33±0.60	-	-	-
F8	Best Fitness	2.37E+04±4.11E+03	2.65E+04±7.20E+03	4.58E+04±3.64E+03	4.28E+04±4.02E+03	6.45E+04±7.43E+03	3.56E+04±3.29E+03
	Mean Fitness	2.76E+04±1.05E+04	3.14E+04±1.08E+04	8.40E+04±6.14E+04	-	-	-
	Peaks Found	7.16±0.37	6.82±0.39	1.80±1.24	-	-	-

7 Conclusion and Future Work

This article has presented a density-assisted framework for EAs, called DAEA, to address DMMOPs. We have employed KDE to construct the density landscape of the concurrently active and/or historically extinct sub-populations and designed the density-assisted search and restart, which combined with density-assisted initialization constitutes the complete DAEA framework. The proposed DAEA could efficiently search and track multiple high-quality solutions and maintain the global diversity of the population, crucial for exploring and adapting to dynamically changing environments. Then, we integrated the LSTM method and memory strategy to predict and respond to environmental changes, reducing the fitness evaluations and expediting the optimization process. The performance of DAEA has been benchmarked on the CEC'2022 dynamic multimodal optimization benchmark suite, and the experimental results have demonstrated that the proposed DAEA is quite competitive.

In future work, we will focus on refining the proposed strategies, including conducting theoretical analyses to further investigate algorithmic performance and exploring their applicability to a broader range of dynamic optimization problems, such as constrained and robust optimization [Xu et al., 2021a; Yazdani et al., 2023a]. Additionally, we will extend the application of DAEA to more general combinatorial and continuous DMMOPs derived from real-world scenarios [Ji et al., 2017; Zheng and Geroliminis, 2016]. These applications will consider not only the global optima but also acceptable local optimal solutions to enhance practical utility. Furthermore, we plan to integrate a dynamic environmental change detection mechanism that activates adaptive response strategies when significant environmental changes are detected.

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